

Guidance to Individual Sustainability Requirements Elicitation

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1 Key Concepts

1.1 Individual Dimension: Definitions

As defined in the Karlskrona manifesto, individual sustainability refers to the maintenance of the private goods of individual human capital. The human capital consists of health, education, skills, knowledge, leadership, and access to services. Individual dimension covers individual freedom, agency, human dignity, and fulfillment.

Individual sustainability refers to longer-term changes in individuals' attitudes, aptitudes and/or behavior that result from their involvement with a particular sports development program or activity. This is achieved by evaluating the consumption output using a consumer's personal and economic well-being as standards. Individual sustainability relies on a capacity for reflection and an awareness of one's past, present, and future to achieve positive living and psychological well-being. It is a person's ability to live a healthy lifestyle that includes creating harmony, interconnections, community, ethically responsible systems, and relatively high levels of awareness in one's values, thoughts, and behaviors.

The individual dimension covers individual freedom and agency (the ability to act in an environment), human dignity and fulfillment. It includes the ability of individuals to thrive, exercise their rights and develop freely.

Individual sustainability includes possessing a well-developed and demonstrated value system that acknowledges the importance and interconnectedness of all global biological and social systems, and our appropriate place within them.

1.2 Human Abilities or Values Pertaining to Individual Sustainability

Individual Sustainability Requirements (ISR) are system-specific requirements derived from underlying human values and human abilities. Human values represent abstract principles that capture what individuals consider important, and

they play a central role in shaping decision-making and behavior. Values function as guiding criteria through which individuals justify actions, make choices, and evaluate themselves, others, and events. These values may include aspects such as achievement, prestige, and wealth, which, although not strictly ethical in nature, significantly influence behavior and decisions. For example, when Google employees urged their CEO to withdraw from Project Maven, their arguments extended beyond moral responsibility to include concerns about potential damage to the company’s reputation and its ability to attract talent. This illustrates that values such as market reputation and public image—while not traditionally categorized as ethical—are nonetheless influential and actively guide decision-making.

Although values are organized in a universal and structured manner, individuals and groups differ substantially in the relative importance they assign to different values. Building on this, Maio et al. conceptualize values as mental representations operating across three interconnected levels: (1) L1 (Value system level), which captures patterns and relationships among values held by individuals; (2) L2 (Abstract interpretation level), where individuals interpret what a value means in a given context (e.g., what constitutes “high-quality software” as an expression of achievement); and (3) L3 (Behavioral level), where values are instantiated through concrete actions and system behaviors. For instance, a search engine company that prioritizes privacy may operationalize this value by ensuring that user queries are never logged.

In this context, operationalizing human values in software refers to the process of identifying relevant values and translating them into concrete, implementable, and measurable constructs that can be validated and verified within a system. ISR serves as this operational bridge by converting abstract values into system-level requirements.

An important empirical finding is that value hierarchies are context-dependent and dynamic. Even for the same individual, the relative importance of values may shift across domains. While some values, such as security or achievement, tend to remain stable, others may change in significance or become irrelevant depending on the application context. This highlights the need for domain-specific elicitation and operationalization of values when deriving ISR.

1.2.1 Unified Characteristics of a Self-Actualized / Fully Functioning Individual

The following characteristics synthesize and de-duplicate the attributes proposed by Maslow and Rogers, providing a consolidated view of a self-actualized or fully functioning individual.

1. Realistic and Accurate Perception

Ability to perceive reality objectively, without distortion from biases, fears, or social conditioning.

2. Acceptance of Self, Others, and Nature

Comfort with oneself and others without excessive guilt or judgment, along

with acceptance of natural processes.

3. **Openness to Experience**
Willingness to engage with new ideas, emotions, and experiences without defensiveness.
4. **Authenticity and Inner Congruence**
Alignment between internal feelings and external behavior; acting in ways true to oneself.
5. **Spontaneity and Naturalness**
Ability to act freely and genuinely without rigid adherence to social expectations.
6. **Problem-Centered Orientation**
Focus on meaningful goals beyond self-interest, with a commitment to solving real-world or societal problems.
7. **Autonomy and Inner Authority**
Independence from external pressures, with decisions guided by internal values rather than societal norms.
8. **Resistance to Social Conditioning**
Critical stance toward rigid norms and institutions when they conflict with personal values.
9. **Desire for Privacy and Reflection**
Capacity for solitude, self-reflection, and independence from constant social interaction.
10. **Deep Interpersonal Relationships and Intimacy**
Ability to form meaningful, empathetic, and authentic relationships with others.
11. **Empathy and Identification with Humanity**
Strong sense of connection, compassion, and responsibility toward others and society.
12. **Democratic and Egalitarian Attitude**
Respect for all individuals and valuing equality and fairness in interactions.
13. **Creativity and Expressiveness**
Ability to think innovatively and express oneself in original and adaptive ways.
14. **Freshness of Appreciation**
Capacity to experience everyday events with renewed wonder and emotional richness.

15. **Peak Experiences**
Experiencing intense moments of joy, insight, or transcendence that provide meaning and fulfillment.
16. **Holistic and Process-Oriented Self**
Viewing life as a continuous evolving process and embracing growth and change.
17. **Value System Transformation**
Shift toward intrinsic values such as growth, meaning, ethics, and self-development.
18. **Reduced Materialism**
Lower emphasis on material possessions, prioritizing experiences and relationships.
19. **Spiritual or Existential Awareness**
Interest in deeper meaning, purpose, or connection beyond the self.
20. **Balanced View of Science and Technology**
Critical yet reflective stance toward technology, recognizing both benefits and limitations.
21. **Ecological and Nature-Oriented Awareness**
Respect and sensitivity toward nature and environmental interconnectedness.

Note: These characteristics are synthesized by merging overlapping concepts from Maslow's self-actualization theory and Rogers' fully functioning person, ensuring removal of redundancy while preserving conceptual completeness.

1.2.2 Schwartz basic human values

It considers the following concepts:-

1. Self-Direction: Independent thought and action-choosing, creating and exploring
2. Simulation: Excitement, novelty, challenging in life
3. Hedonism: Pleasure or sensuous gratification for oneself
4. Achievement: Personal success through demonstrating competence
5. Power: social status and prestige, control or dominance over people and resources
6. Security: safety, harmony and stability of society or self
7. Conformity: restraint of actions, inclinations or impulses

8. Tradition: respect, commitment and acceptance of the customs and ideas
9. Benevolence: preserving and enhancing the welfare
10. Universalism: understanding, appreciation, tolerance and protection for the welfare

Value Theme	Description in Software Context	Mapped Schwartz Value
Conformity	The system supports adherence to established rules, standards, and expected norms of behaviour, ensuring that users act within predefined constraints and guidelines.	Conformity – Rules
Pleasure	The system provides enjoyment, satisfaction, or a positive emotional experience during interaction, contributing to user engagement and comfort.	Hedonism
Dignity	The system ensures that users are treated with respect and are not exposed to humiliation, embarrassment, or loss of self-worth during interaction.	Face
Inclusiveness	The system accommodates diverse user groups, including differences in language, culture, ability, and background, ensuring equitable participation.	Universalism – Concern
Sense of Belonging	The system enables users to feel socially connected, included, and part of a community through interaction and communication features.	Security – Personal
Freedom	The system provides users with the ability to make choices, control actions, and avoid unnecessary restrictions in interaction.	Self-direction – Action
Independence	The system allows users to perform tasks without reliance on external systems, intermediaries, or additional support, enhancing autonomy.	Self-direction – Action
Wealth	The system reflects or impacts financial considerations, such as cost savings, monetary benefits, or resource ownership.	Power – Resources
Privacy	The system protects user data, personal information, and identity from unauthorized access, exposure, or misuse.	Self-direction – Action
Security	The system safeguards users from threats, vulnerabilities, and risks, ensuring safe and protected interaction.	Security – Personal

Table 1: Human Value Themes and Their Mapping to Schwartz’s Model

Value Theme	Description in Software Context	Possible Mapping to Schwartz Value
Trust	The system is perceived as reliable, safe, and dependable by users, fostering confidence in its behaviour and outputs.	Conformity – Interpersonal
Correctness	The system produces accurate, expected, and logically consistent outputs, ensuring correctness of functionality.	No clear direct mapping
Compatibility	The system operates effectively across different devices, platforms, and environments without conflicts or limitations.	Power – Resources
Portability	The system allows users to transfer, migrate, or reuse data and functionalities across different contexts and devices.	Self-direction – Action / Power – Resources
Reliability	The system maintains stable performance over time, minimizing failures, crashes, and unexpected behaviour.	No clear direct mapping
Efficiency	The system enables users to complete tasks with minimal time, effort, and resource consumption.	Power – Resources
Energy Preservation	The system optimizes energy consumption, such as reducing battery usage or computational overhead.	Power – Resources
Usability	The system is easy to learn, understand, and operate, enabling users to perform tasks effectively and with minimal difficulty.	Benevolence – Caring
Accessibility	The system ensures that users with varying abilities, including disabilities, can effectively interact with and benefit from its functionalities.	Benevolence – Caring / Universalism – Concern
Longevity	The system remains functional, maintainable, and usable over a long period, supporting long-term user needs and stability.	Security – Personal

Table 2: System Value Themes and Their Interpreted Mapping to Schwartz’s Model

1.3 SuSAF Framework for Individual Sustainability

The *SusAF* approach proposes the following categories of sustainability dimensions-

Health

Health means the state of a person’s mental or physical condition.

- How can the product or service improve or worsen a person’s physical, mental, and/or emotional health?

- For example, can it make a person feel anything good or bad – e.g., (un)valued, (dis)respected, (in)dependent, or coerced?

Lifelong Learning

Lifelong learning means the use of learning opportunities throughout people's lives for continuous development.

- How can the product or service affect people's competencies?

Privacy

Privacy means being free from intrusion or disturbance in one's private life.

- How can the product or service expose (or help to hide) a person's identity, whereabouts, or relations?

Safety

Safety means being protected from danger, risk, or injury.

- How can the product or service expose (or protect) a person from physical harm?
- How can it make a person feel more (or less) exposed to harm?
- What if the product or service is used in an unintended way?

Self-awareness and Free Will

Self-awareness and free will mean the capacity of an individual to act or make decisions on their own.

- How can the product or service empower (or prevent) a person from taking an action or decision when necessary?
- Can those affected by the product or service understand its implications, express concerns, or be represented by someone?

1.4 Human Values and Artificial Intelligence (AI)

As AI-driven systems grow more sophisticated and autonomous, they raise fundamental questions about alignment: not just whether the underlying AI models behave safely, but whether the software systems that embed and orchestrate them uphold human values like trust, honesty, autonomy, fairness, and accountability.

One of the central challenges in addressing these values lies in the highly subjective prioritization of values. This is especially significant for AI systems due to their autonomy, opacity, and wide-reaching societal impact. Different

stakeholders (e.g., developers, users, regulators, domain experts) and impacted communities often hold divergent or conflicting value priorities. For example, a development team may prioritize system efficiency and scalability, while end users may emphasize transparency in decision-making, equity, or preservation of autonomy. Values such as sustainability, privacy, accountability, and inclusiveness are often interpreted and weighted differently, leading to miscommunication, overlooked needs, or unintended harms. Conflicts may arise within user groups as well. For example, law enforcement may emphasize national security, while civil rights groups prioritize privacy protections. Similarly, designers might face a trade-off between creating highly secure systems and maintaining ease of use for end users.

If these differences remain unrecognized or unresolved, the resulting design decisions may not align with stakeholder expectations, undermining trust in AI systems and potentially causing real-world consequences.

Value alignment in AI-driven systems requires translating abstract human values into measurable indicators, embedding them into system architecture, continuously monitoring their adherence, and adapting them dynamically across contexts and stakeholders.

Self-adaptive socio-technical systems (SASTSs)—especially those driven by AI—raises serious concerns about the erosion of human autonomy, values, and ethical control. While such systems optimize decisions automatically, they often suffer from misaligned objectives, algorithmic bias, and neglect of human values, particularly “soft ethics” such as personal preferences and cultural norms. The authors emphasize that current regulatory approaches (e.g., GDPR) are largely reactive and insufficient, and that systems must instead be designed to actively empower humans during interaction, rather than treating them as passive users